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FURTHER READING

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311 Chronic Kidney Disease

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Chronic kidney disease (CKD) encompasses a spectrum of pathophysiologic processes associated with abnormal kidney function, often with a progressive decline in glomerular filtration rate (GFR). The risk of worsening CKD is closely linked to both the GFR and the amount of albuminuria. **Figure 311-1** provides a staging of CKD stratified by the estimates for further progressive decline of GFR based on these two parameters.

The dispiriting term *end-stage renal disease* represents a stage of CKD where the accumulation of toxins, fluid, and electrolytes normally excreted by the kidneys leads to death unless the toxins are removed by renal replacement therapy, using dialysis or kidney transplantation. These interventions are discussed in **Chaps. 312** and **313**. End-stage renal disease will be supplanted in this chapter by the term stage 5 CKD.

PATHOPHYSIOLOGY OF CKD

The pathophysiology of CKD involves two broad mechanisms of damage: (1) specific initiating mechanisms particular to the underlying etiology (e.g., genetic abnormalities in kidney development, immune complex deposition, and inflammation in certain types of glomerulonephritis, or toxin exposure in certain diseases of the renal tubules and interstitium), and (2) nonspecific mechanisms involving hyperfiltration and hypertrophy of the remaining viable nephrons, which are common consequences of long-term reduction of renal mass, irrespective of underlying etiology. The responses to reduction in nephron number are mediated by vasoactive hormones, cytokines, and growth factors. Eventually, the short-term adaptations of hyperfiltration and hypertrophy to maintain GFR become maladaptive as the increased pressure and flow within the nephron predisposes to distortion of glomerular architecture, abnormal podocyte function, and disruption of the filtration barrier, leading to sclerosis and dropout of the remaining nephrons **(Fig. 311-2)**. Increased intrarenal activity of the renin-angiotensin system (RAS) appears to contribute both to the initial compensatory hyperfiltration and to the subsequent maladaptive hypertrophy and sclerosis. This process explains why a reduction in renal mass from an isolated insult may lead to a progressive decline in renal function over many years and the efficacy of pharmacologic approaches that attenuate this response **(Fig. 311-3)**.

IDENTIFICATION OF RISK FACTORS AND STAGING OF CKD

There has been significant recent progress in the identification of risk factors that increase the risk for CKD, even in individuals with normal GFR **(Table 311-1)**.

Adults with such risk factors should be monitored at least every 2 years for albuminuria, decline in eGFR, and blood pressure abnormalities, so that a clinical management pathway can be planned.

Most recently identified risk factors for which there is now a consensus include a past episode of apparently recovered acute kidney injury (AKI), tobacco use, and many forms of apparently resolved childhood and adolescent kidney disease. There is also an increasing awareness of the role of genetic risk factors, which account for 15–40% of adult-onset CKD, with the percentage often depending on the contribution of demographic structure and history to the genetic variation for any population. Many rare inherited forms of CKD follow a Mendelian inheritance pattern, often as part of a systemic syndrome, with the most

Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012

GFR categories (mL/min/1.73 m²) description and range				Persistent albuminuria categories description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–300 mg/g 3–30 mg/mmol	>300 mg/g >30 mg/mmol
	G1	Normal or high	≥90			
	G2	Mildly decreased	60–89			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	<15			

FIGURE 311-1 Kidney Disease Improving Global Outcome (KDIGO) classification of chronic kidney disease (CKD). Gradation of color from green to red corresponds to increasing risk and progression of CKD. GFR, glomerular filtration rate. *(Reproduced with permission from KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney Int Suppl 3:5, 2013.)*

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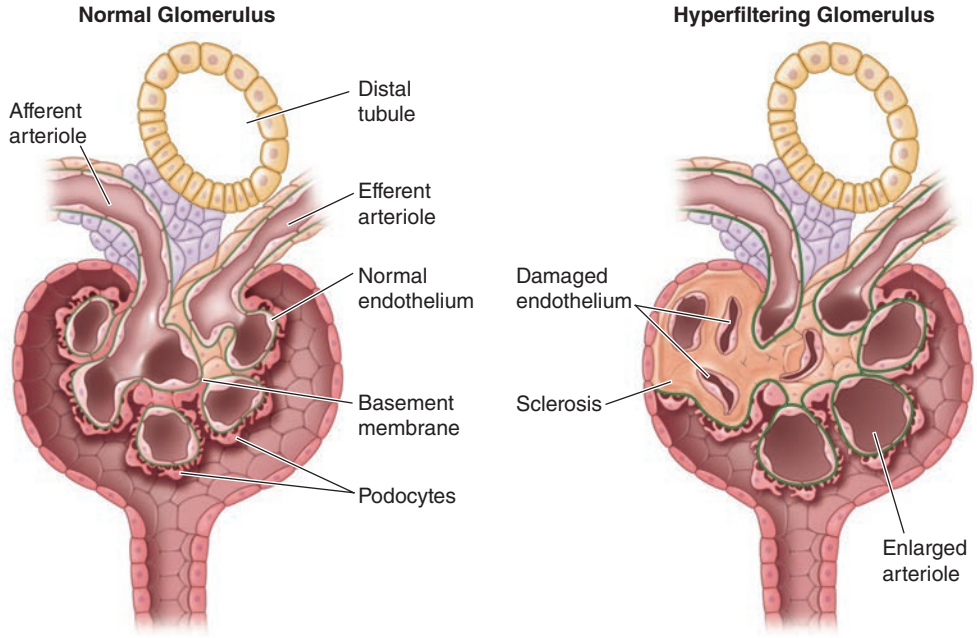


FIGURE 311-2 ***Left:*** Schema of the normal glomerular architecture. ***Right:*** Secondary glomerular changes associated with a reduction in nephron number, including enlargement of capillary lumens and focal adhesions, which are thought to occur consequent to compensatory hyperfiltration and hypertrophy in the remaining nephrons. *(From JR Ingelfinger: Is microanatomy destiny?. N Engl J Med 348:99, 2003. Copyright © 2003, Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)*

common in this category being autosomal dominant polycystic kidney disease (ADPKD). In addition, it is now appreciated that many unique, kindred-specific, site-specific copy number variants and microdeletions, as well as functional variants at >60 genetic loci known to harbor systemic and kidney-only disease pathogenic mutations, also contribute to risk for pleiotropic presentations of CKD (Table 311-2). Many of the genes with identified CKD-causing mutations are expressed in the podocytes of the renal glomeruli or in the glomerular basement membrane, but others are expressed in tubule segments with a primary tubulointerstitial process and secondary glomerular injury. Given the significant contribution of monogenic disease etiologies, consideration is now given to chromosomal microarray and genome or exome sequencing for CKD of unknown cause in young adults, as noted below under Evaluation and Management of Patients with CKD.

In addition, recent research in the genetics of predisposition to common complex diseases has revealed DNA sequence variants at a

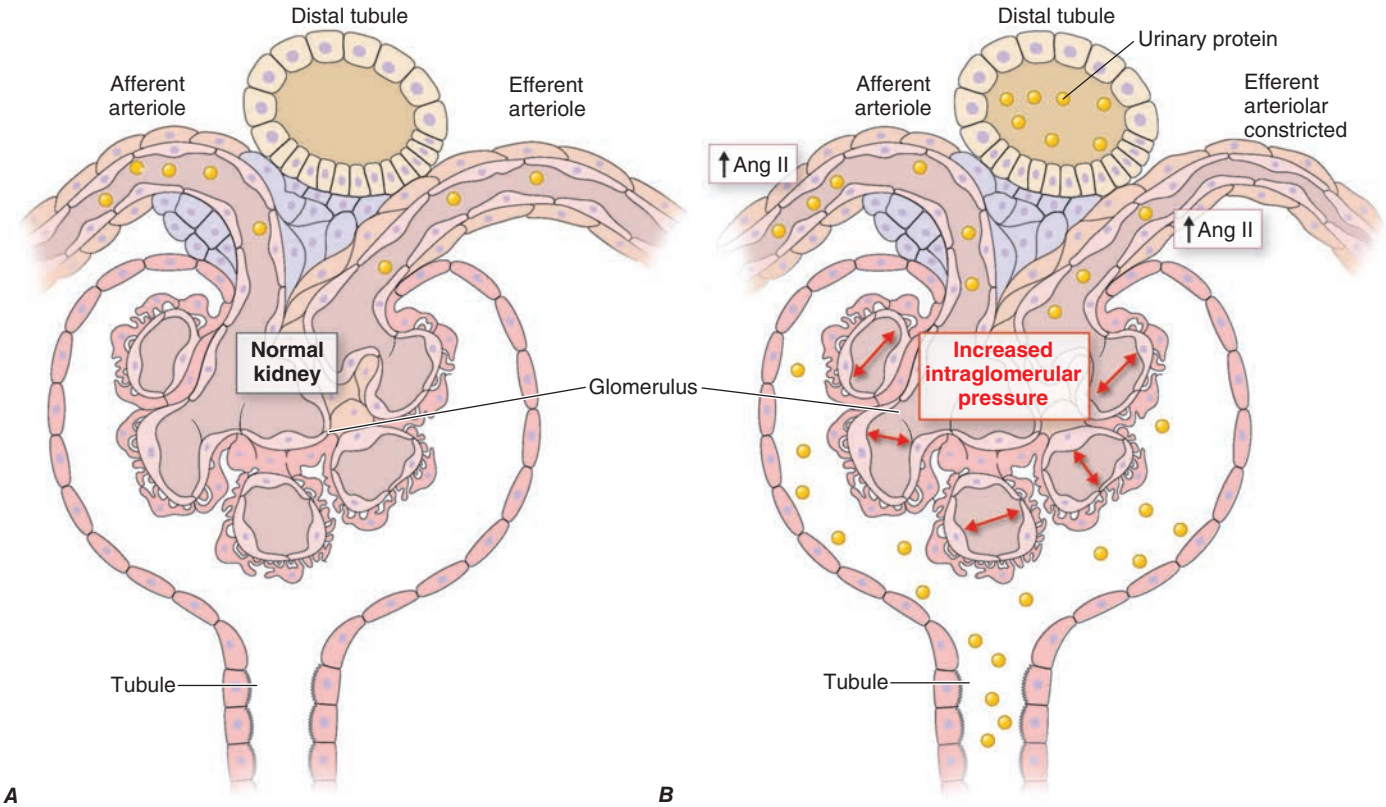


FIGURE 311-3 Schematic representation of the effect of intraglomerular hypertension on nephron survival.

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TABLE 311-1 Risk Factors for Chronic Kidney Disease (CKD) in Adulthood by Category[a]

Chronic Nonrenal (Systemic) Disease
Diabetes and metabolic syndrome
Autoinflammatory disease (e.g., lupus, vasculitis, cancer immunotherapy)
Infections (e.g., HIV, HBV, HCV)
Absence of infection (JCV)
Nephrotoxic exposure (including many antineoplastic therapies)
Hypertension (risk, cause, or consequence)
Demographic, Anthropomorphic, Ancestry, Geographic
Age
Sex
Population ancestry
Family history
Region-specific CKD risk of uncertain etiology (e.g., Central America, Sri Lanka, and indigenous peoples of Australia and New Zealand)
Childhood and Adolescent States and Diseases
Premature and SGA birth
Increased BMI
Persistent asymptomatic microscopic hematuria
Elevated blood pressure
Childhood kidney disease (even resolved)
Treated childhood cancer
Adult Onset
Prior acute kidney injury
Preeclampsia
Kidney donation (or other acquired nephrectomy)
Genetic
Monogenic Mendelian inheritance
Polygenic complex inheritance
Viral Infection
HIV infection (HIVAN)
SARS-CoV-2 (COVAN)
Lifestyle
Smoking
Diet
Physical activity

[a]Not biomarkers.

Abbreviations: BMI, body mass index; COVAN, COVID-19–associated nephropathy; HBV, hepatitis B virus; HCV, hepatitis C virus; HIVAN, HIV-associated nephropathy; JCV, JC virus; SGA, small for gestational age.

TABLE 311-2 Monogenic Risk Loci for Chronic Kidney Disease

Copy number variants
17q12
22q11.2
16p11.2
Single nucleotide variants at four most predominant genetic loci with mendelian inheritance
Genes for autosomal dominant polycystic kidney disease
ADPKD1
ADPKD2
GANAB
DNAJBII
ALG9
Genes for type IV collagen-associated nephropathy
COL4A3
COL4A4
COL4A5
Genes for autosomal dominant tubulointerstitial kidney disease
UMOD
MUCI
REN
HNFIB
SEC6IAI
Genes with known common variants that confer increased risk with odds ratio exceeding 2 with non-Mendelian inheritance patterns
APOL1

number of genetic loci that are associated with common forms of CKD. A striking example is the finding of allelic versions of the *APOL1* gene, of West African population ancestry, which contributes to the several-fold higher frequency of certain common etiologies of nondiabetic CKD (e.g., focal segmental glomerulosclerosis) observed among African and Hispanic Americans, in major regions of continental Africa and the global African diaspora. The prevalence in West African populations seems to have arisen as an evolutionary adaptation conferring protection from tropical pathogens. As in other common diseases with a heritable component, environmental triggers (e.g., a viral pathogen) transform genetic risk into disease.

To stage CKD, it is necessary to estimate the GFR rather than relying on serum creatinine concentration (Table 311-3). Many laboratories now report an estimated GFR, or eGFR, using one of these equations. These equations are valid only if the patient is in steady state, that is, the serum creatinine is neither rising nor falling over days. The societal implications of adjustment for a construct of race have been the subject of important recent discourse, with the idea that more individually sound adjustments without potentially negative racial categorizations be developed.

The normal annual mean decline in GFR with age from the peak GFR (~120 mL/min per 1.73 m^2) attained during the third decade of life is ~1 mL/min per year per 1.73 m^2, reaching a mean value of 70 mL/min per 1.73 m^2 at age 70, with considerable interindividual variability. Although reduced GFR is expected with aging, the lower GFR signifies a true loss of kidney function with attendant consequences in terms of risk of CKD complications and requirement for dose adjustment of medications. The mean GFR is lower in women than in men. For example, a woman in her eighties with a laboratory report of serum creatinine in the normal range may have a GFR of <50 mL/min per 1.73 m^2. Relatedly, even a mild elevation in serum creatinine concentration often signifies a substantial reduction in GFR in older individuals.

Measurement of albuminuria is also helpful for monitoring nephron injury and the response to therapy in many forms of CKD, especially chronic glomerular diseases. The cumbersome 24-h urine collection has been replaced by measurement of urinary albumin-to-creatinine ratio (UACR) in one and preferably several spot first-morning urine samples as a measure pointing to glomerular injury. Even in patients with negative conventional urinary dipstick tests for protein, persistent UACR >2.5 mg/mmol (male) or >3.5 mg/mmol (female) on two to three occasions serves as a marker not only for early detection of primary kidney disease but for systemic microvascular disease as well.

TABLE 311-3 Recommended Equations for Estimation of Glomerular Filtration Rate (GFR) Using Serum Creatinine Concentration (S_{CR}), Age, Sex, Race, and Body Weight

1. Equation from the Modification of Diet in Renal Disease Study
Estimated GFR (mL/min per 1.73 m^2) = $1.86 \times (S_{Cr})^{-1.154} \times (age)^{-0.203}$
Multiply by 0.742 for women
Multiply by 1.21 for African ancestry (currently under review)
2. CKD-EPI Equation
GFR = $141 \times \min(S_{Cr}/\kappa, 1)^{\alpha} \times \max(S_{Cr}/\kappa, 1)^{-1.209} \times 0.993^{Age}$
Multiply by 1.018 for women
Multiply by 1.159 for African ancestry (currently under review)
where S_{Cr} is serum creatinine in mg/dL, κ is 0.7 for females and 0.9 for males, α is –0.329 for females and –0.411 for males, min indicates the minimum of S_{Cr}/κ or 1, and max indicates the maximum of S_{Cr}/κ or 1.

Abbreviation: CKD-EPI, Chronic Kidney Disease Epidemiology Collaboration.

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A Kidney Failure Risk (KFR) equation has been devised to predict the risk of progression to stage 5 dialysis-dependent kidney disease. The equation is available on many sites online (for example, *www.kidneyfailurerisk.com*) and uses age, sex, region (North American or non–North American), GFR, and UACR. It has been validated in several cohorts around the world, although the risk for progression appears to be greater in North America, accounting for the regional adjustment in the equation.

Stages 1 and 2 CKD are usually asymptomatic, such that the recognition of CKD occurs more often as a result of laboratory testing in clinical settings other than suspicion of kidney disease. Moreover, in the absence of the risk factors noted above, population-wide screening is not recommended. With progression to CKD stages 3 and 4, clinical and laboratory complications become more prominent. Virtually all organ systems are affected, but the most evident complications include anemia with easy fatigability; decreased appetite with progressive malnutrition; abnormalities in calcium, phosphorus, and mineral-regulating hormones, such as $1,25(OH)_2D_3$ (calcitriol), parathyroid hormone (PTH), and fibroblast growth factor 23 (FGF-23); and abnormalities in sodium, potassium, water, and acid-base homeostasis. Many patients, especially older individuals, will have eGFR values compatible with stage 2 or 3 CKD. However, the majority of these patients will show no further deterioration of kidney function. In this setting, it is advised to recheck kidney function, and if it is stable and not associated with proteinuria, the patient can usually be followed with interval repeat testing without referral to a nephrologist. However, caution should be exercised in terms of potential exposure to potential nephrotoxins or interventions that risk AKI and also with respect to medication dose adjustment. If repeat testing shows declining GFR, albuminuria, or uncontrolled hypertension, referral to a nephrologist is appropriate. If the patient progresses to stage 5 CKD (GFR <15 mL/min), toxins accumulate such that patients usually experience a disturbance in their activities of daily living, well-being, nutritional status, and water and electrolyte homeostasis, eventuating in the uremic syndrome.

ETIOLOGY AND EPIDEMIOLOGY

It has been estimated from population data that at least 6% of the adult population in the United States has CKD at stages 1 and 2. An additional 4.5% of the U.S. population is estimated to have stages 3 and 4 CKD. **Table 311-4** lists the five most frequent categories of causes of CKD, cumulatively accounting for >90% of the CKD disease burden worldwide. The relative contribution of each category varies among different geographic regions. The most frequent cause of CKD in North America and Europe is diabetic nephropathy, most often secondary to type 2 diabetes mellitus. Patients with newly diagnosed CKD often have hypertension. When no overt evidence for a primary glomerular or tubulointerstitial kidney disease process is present, CKD is frequently attributed to hypertension. However, it is now appreciated that some of these patients may have a subclinical primary glomerulopathy, such as focal segmental or global glomerulosclerosis. In other patients, progressive nephrosclerosis and hypertension are the renal correlates of a systemic vascular disease, often also involving large and small vessels elsewhere, such as the heart and brain. This latter combination is especially common in older patients, among whom chronic kidney ischemia as a cause of CKD may be underdiagnosed.

TABLE 311-4 Leading Categories of Etiologies of Chronic Kidney Disease (CKD)[a]

- Diabetic nephropathy
- Glomerulonephritis
- Hypertension-associated CKD (includes vascular and ischemic kidney disease and primary glomerular disease with associated hypertension)
- Autosomal dominant polycystic kidney disease
- Other cystic and tubulointerstitial nephropathy

[a]Relative contribution of each category varies with geographic region and race.

PATHOPHYSIOLOGY AND BIOCHEMISTRY OF UREMIA

Although serum urea and creatinine concentrations are used to measure the excretory capacity of the kidneys, accumulation of these two molecules themselves does not account for the symptoms and signs that characterize the uremic syndrome in advanced CKD. Large numbers of toxins that accumulate when GFR declines have been implicated in the uremic syndrome. These include water-soluble, hydrophobic, protein-bound, charged, and uncharged nitrogen-containing nonvolatile products of metabolism. It is thus evident that the serum concentrations of urea and creatinine should be viewed as being readily measured but very incomplete surrogate markers for retained toxins, and monitoring the levels of urea and creatinine in the patient with impaired kidney function represents a vast oversimplification of the uremic state.

The uremic syndrome involves more than renal excretory failure. A host of metabolic and endocrine functions normally performed by the kidneys are also impaired and can result in anemia, malnutrition, and abnormal metabolism of carbohydrates, fats, and proteins. Furthermore, plasma levels of many hormones, including PTH, FGF-23, insulin, glucagon, steroid hormones including vitamin D and sex hormones, and prolactin change with CKD as a result of reduced excretion, decreased degradation, or abnormal regulation. Finally, CKD is associated with increased systemic inflammation. Elevated levels of C-reactive protein are detected along with other acute-phase reactants, whereas levels of so-called negative acute-phase reactants, such as albumin and fetuin, decline. Thus, the inflammation associated with CKD is important in the malnutrition-inflammation-atherosclerosis/calcification syndrome, which contributes in turn to the acceleration of vascular disease and morbidity associated with advanced kidney disease.

In summary, the pathophysiology of the uremic syndrome can be divided into manifestations in three spheres of dysfunction: (1) those consequent to the accumulation of toxins that normally undergo renal excretion; (2) those consequent to the loss of other kidney functions, such as fluid and electrolyte homeostasis and hormone regulation; and (3) progressive systemic inflammation and its vascular and nutritional consequences.

CLINICAL AND LABORATORY MANIFESTATIONS OF CKD AND UREMIA

Uremia leads to disturbances in the function of virtually every organ system. Chronic dialysis can reduce the incidence and severity of many of these disturbances, so that the florid manifestations of uremia have largely disappeared in the modern health care setting. However, even optimal dialysis therapy is not completely effective as renal replacement therapy because some disturbances resulting from impaired kidney function fail to respond to dialysis.

FLUID, ELECTROLYTE, AND ACID-BASE DISORDERS

Sodium and Water Homeostasis With normal kidney function, tubular excretion of filtered sodium and water matches intake. Many forms of kidney disease disrupt this balance such that dietary intake of sodium exceeds its excretion, leading to sodium retention and attendant extracellular fluid volume (ECFV) expansion. This expansion may contribute to hypertension, which itself can accelerate nephron hyperfiltration and injury. As long as water intake does not exceed the capacity for renal water clearance, the ECFV expansion will be isonatric and the patient will have a normal plasma sodium concentration. Hyponatremia is not commonly seen in CKD patients but, when present, often responds to water restriction. The patient with ECFV expansion (peripheral edema, sometimes hypertension poorly responsive to therapy) should be counseled regarding salt restriction. Thiazide diuretics have limited utility in stages 3–5 CKD, such that administration of loop diuretics, including furosemide, bumetanide, or torsemide, may be needed. Resistance to loop diuretics in CKD often mandates use of higher doses than those used in patients with higher GFR. The combination of loop diuretics with metolazone may

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be helpful. Diuretic resistance with intractable edema and hypertension in advanced CKD may serve as an indication to initiate dialysis.

Rarely, patients with CKD may have impaired renal conservation of sodium and water. When an extrarenal cause for fluid loss, such as gastrointestinal (GI) loss, is present, these patients may be prone to ECFV depletion because of the inability of the failing kidney to reclaim filtered sodium adequately. Any depletion of ECFV, whether due to GI losses, renal sodium loss, or overzealous diuretic therapy, can further compromise kidney function through underperfusion, or a "prerenal" state, leading to acute-on-chronic kidney failure. In this setting, holding or adjusting the diuretic dose or even cautious volume repletion with normal saline may return the ECFV to normal and restore renal function to baseline.

Potassium Homeostasis In CKD, the decline in GFR is not necessarily accompanied by a parallel decline in urinary potassium excretion, which is predominantly mediated by aldosterone-dependent secretion in the distal nephron. Another defense against potassium retention in these patients is augmented potassium excretion in the GI tract. Notwithstanding these two homeostatic responses, hyperkalemia may be precipitated in certain settings. These include increased dietary potassium intake, hemolysis, transfusion of stored red blood cells, and metabolic acidosis. Importantly, a host of medications can inhibit renal potassium excretion and lead to hyperkalemia. The most important medications in this respect include the RAS inhibitors and spironolactone and other potassium-sparing diuretics such as amiloride, eplerenone, and triamterene. The benefits of the RAS inhibitors in ameliorating hyperfiltration and progression of CKD often favor their cautious and judicious use with very close monitoring of plasma potassium concentration. Coadministration of potassium-lowering agents such as patiromer may allow for the use of RAS inhibitors with reduced risk of hyperkalemia.

Certain causes of CKD can be associated with earlier and more severe disruption of potassium secretory mechanisms in the distal nephron, out of proportion to the decline in GFR. These include conditions associated with hyporeninemic hypoaldosteronism, such as diabetes, and renal diseases that preferentially affect the distal nephron, such as obstructive uropathy and sickle cell nephropathy.

Hypokalemia is not common in CKD and usually reflects markedly reduced dietary potassium intake, especially in association with excessive diuretic therapy or concurrent GI losses. The use of potassium supplements and potassium-sparing diuretics may be risky in patients with impaired renal function and needs to be monitored closely.

Metabolic Acidosis Metabolic acidosis is a common disturbance in CKD. The majority of patients can still acidify the urine, but they produce less ammonia and, therefore, cannot excrete the quantity of protons required to maintain acid-base balance in most diets. Hyperkalemia, if present, further depresses ammonia production. The combination of hyperkalemia and hyperchloremic metabolic acidosis is often present, even at earlier stages of CKD, in patients with diabetic nephropathy or in those with predominant tubulointerstitial disease or obstructive uropathy. With further declining GFR, the total urinary net daily acid excretion may be severely limited to less than 30–40 mmol, and the accumulation of anions of retained organic acids can then lead to an anion-gap metabolic acidosis. Thus, the non-anion-gap metabolic acidosis seen in earlier stages of CKD may be complicated by the addition of an anion-gap metabolic acidosis as CKD progresses. In most patients, the metabolic acidosis is mild; the pH is rarely <7.32 and can usually be corrected with oral sodium bicarbonate supplementation. Studies have suggested that even modest degrees of metabolic acidosis may be associated with the development of protein catabolism and progression of CKD.

TREATMENT

Fluid, Electrolyte, and Acid-Base Disorders

Dietary salt restriction and the use of loop diuretics, occasionally in combination with metolazone, may be needed to maintain euvolemia. Water restriction is indicated only if there is hyponatremia.

Hyperkalemia often responds to dietary restriction of potassium, the use of kaliuretic diuretics, and both avoidance of potassium supplements (including occult sources, such as dietary salt substitutes) and dose reduction or avoidance of potassium-retaining medications (especially RAS inhibitors). Kaliuretic diuretics promote urinary potassium excretion, whereas potassium-binding resins, such as calcium resonium, sodium polystyrene, or patiromer, can promote potassium loss through the GI tract and may reduce the incidence of hyperkalemia. Intractable hyperkalemia is an indication (although uncommon) to consider institution of dialysis in a CKD patient. The renal tubular acidosis and subsequent anion-gap metabolic acidosis in progressive CKD will respond to alkali supplementation, typically with sodium bicarbonate. Recent studies suggest that this replacement should be considered when the serum bicarbonate concentration falls below 20–23 mmol/L to avoid the protein catabolic state seen with even mild degrees of metabolic acidosis and to slow the progression of CKD. The sodium load in sodium bicarbonate supplementation needs to be taken into account, when ECFV expansion is present.

DISORDERS OF CALCIUM AND PHOSPHATE METABOLISM

The principal complications of abnormalities of calcium and phosphate metabolism in CKD occur in the skeleton and the vascular bed, with occasional involvement of soft tissues. It is likely that disorders of bone turnover and disorders of vascular and soft tissue calcification are related to each other.

Bone Manifestations of CKD The major disorders of bone disease can be classified into those associated with high bone turnover with increased PTH levels (including osteitis fibrosa cystica, the classic lesion of secondary hyperparathyroidism), osteomalacia due to reduced effect of the active forms of vitamin D, and low bone turnover with low or normal PTH levels (adynamic bone disease) or most often combinations of the foregoing.

The pathophysiology of secondary hyperparathyroidism and the consequent high-turnover bone disease is related to abnormal mineral metabolism through the following series of interrelated mechanisms: (1) declining GFR leads to reduced excretion of phosphate and, thus, phosphate retention; (2) the retained phosphate stimulates increased synthesis of both FGF-23 by osteocytes and of PTH and also stimulates growth of parathyroid gland mass; and (3) PTH production is stimulated by decreased levels of ionized calcium, which, in turn, result from decreased levels of renal calcitriol production with reduced kidney mass and suppression of calcitriol production due to phosphate retention and elevated levels of FGF-23, which also increases degradation of calcitriol. Low calcitriol levels contribute to hyperparathyroidism, both by leading to hypocalcemia and also by a direct effect on PTH gene transcription. In addition, the normal inhibitory effect of FGF-23 on PTH production is Klotho-dependent and is also attenuated in CKD. These changes start to occur when the GFR falls below 60 mL/min, though some studies point to retention of phosphate as an event antedating measurable reduction in GFR, together with early elevation of FGF-23 as well. FGF-23 is part of a family of phosphatonins that promotes phosphate excretion, and high levels of FGF-23 are an independent risk factor for left ventricular hypertrophy and are associated with increased mortality due to several classes of complications in CKD, dialysis, and kidney transplant patients.

Hyperparathyroidism stimulates bone turnover and leads to osteitis fibrosa cystica. Bone histology shows abnormal osteoid, bone and bone marrow fibrosis, and, in advanced stages, the formation of bone cysts, sometimes with hemorrhagic elements so that they appear brown in color; hence, the term *brown tumor*. Clinical manifestations of severe hyperparathyroidism include bone pain and fragility, brown tumors, compression syndromes, and resistance to erythropoiesis-stimulating agents (ESA) in part related to the bone marrow fibrosis. Furthermore, PTH itself is considered a uremic toxin, and high levels are associated

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FIGURE 311-4 Tumoral calcinosis. This patient was on hemodialysis for many years and was nonadherent to dietary phosphorus restriction or the use of phosphate binders. He was chronically severely hyperphosphatemic. He developed an enlarging painful mass on his arm that was extensively calcified.



FIGURE 311-5 Calciphylaxis. This peritoneal dialysis patient was on chronic warfarin therapy for atrial fibrillation. She noticed a small painful nodule on the abdomen that was followed by progressive skin necrosis and ulceration of the anterior abdominal wall. She was treated with hyperbaric oxygen, intravenous thiosulfate, and discontinuation of warfarin, with slow resolution of the ulceration.

with muscle weakness, fibrosis of cardiac muscle, and constitutional symptoms.

Adynamic bone disease is increasing in prevalence, especially among diabetics and older patients. It is characterized by reduced bone volume and mineralization and may result from excessive suppression of PTH production, chronic inflammation, or both. Suppression of PTH can result from the use of vitamin D preparations or from excessive calcium exposure in the form of calcium-containing phosphate binders or high-calcium dialysis solutions.

Complications of adynamic bone disease include an increased incidence of fracture and bone pain and an association with increased vascular and cardiac calcification. Occasionally, the calcium will precipitate in the soft tissues into large concretions termed *tumoral calcinosis* (Fig. 311-4). Patients with adynamic bone disease often experience the most severe symptoms of musculoskeletal pain, owing to the inability to repair the microfractures that occur normally as a part of healthy skeletal homeostasis with regular physical activity. Patients with advanced CKD experience more frequent fractures than their age-matched controls. Osteomalacia is a distinct process, consequent to reduced production and action of $1,25(OH)_2D_3$, leading to accumulation of nonmineralized osteoid.

Calcium, Phosphorus, and the Cardiovascular System There is a strong association between hyperphosphatemia and increased cardiovascular mortality in patients with CKD. Hyperphosphatemia and hypercalcemia are associated with increased vascular calcification, but it is unclear whether the excessive mortality is mediated by this mechanism. Studies using computed tomography (CT) and electron-beam CT scanning show that CKD patients have calcification in the media of coronary arteries and even heart valves that appears to be orders of magnitude greater than that in patients without renal disease. The magnitude of the calcification is proportional to age and hyperphosphatemia and is also associated with low PTH levels and low bone turnover. It is possible that in CKD patients ingested calcium cannot be incorporated into bones with low turnover and, therefore, is deposited at extraosseous sites, such as the vascular bed and soft tissues. There is a similar association between osteoporosis and vascular calcification in the general population. Finally, hyperphosphatemia can induce a change in gene expression in vascular cells to an osteoblast-like profile, leading to vascular calcification and even ossification.

Other Complications of Abnormal Mineral Metabolism Calciphylaxis is a devastating condition seen almost exclusively in patients with advanced CKD. It is heralded by painful livedo reticularis and subcutaneous nodules that advance to patches of ischemic necrosis, especially on the legs, thighs, abdomen, and breasts (Fig. 311-5). Pathologically, there is evidence of vascular occlusion in association with extensive vascular and soft tissue calcification. It appears that this condition is increasing in incidence. Originally it was ascribed to severe abnormalities in calcium and phosphorus control in dialysis patients, usually associated with advanced hyperparathyroidism. However, more recently, calciphylaxis has been seen with increasing frequency in the absence of severe hyperparathyroidism. Warfarin is still used in some dialysis patients in whom direct oral anticoagulants (DOACs) are contraindicated, and one of the effects of warfarin therapy is to decrease the vitamin K–dependent activation of matrix GLA protein. This latter protein is important in preventing vascular calcification. Thus, warfarin treatment is considered a risk factor for calciphylaxis, and if a patient develops this syndrome, this medication should be discontinued and alternative means of anticoagulation should be chosen, depending on the specific underlying indication for anticoagulation.

TREATMENT

Disorders of Calcium and Phosphate Metabolism

The optimal management of secondary hyperparathyroidism and osteitis fibrosa is prevention. Once the parathyroid gland mass is very large, it is difficult to control the disease. Careful attention should be paid to the plasma phosphate concentration in CKD patients, who should be counseled on a low-phosphate diet as well as the appropriate use of phosphate-binding agents, which are taken with meals and complex dietary phosphate to limit its GI absorption. Examples of phosphate binders are calcium acetate and calcium carbonate. A major side effect of calcium-based phosphate binders is calcium accumulation and hypercalcemia, especially in patients with low-turnover bone disease. Sevelamer and lanthanum are non-calcium-containing polymers that also function as phosphate binders; they do not predispose CKD patients to hypercalcemia and may attenuate calcium deposition in the vascular bed. Tenapanor is a sodium-proton inhibitor that decreases GI phosphate absorption and may be useful to manage hyperphosphatemia in CKD and dialysis patients.

Calcitriol exerts a direct suppressive effect on PTH secretion and also indirectly suppresses PTH secretion by raising the concentration of ionized calcium. However, calcitriol therapy may result in hypercalcemia and/or hyperphosphatemia through increased GI absorption of these minerals. Certain analogues of calcitriol are

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available (e.g., paricalcitol) that suppress PTH secretion with less attendant hypercalcemia.

Recognition of the role of the extracellular calcium-sensing receptor has led to the development of calcimimetic agents that enhance the sensitivity of parathyroid cells to the suppressive effect of calcium. This class of drug, which includes cinacalcet and etelcalcetide, produces a dose-dependent reduction in PTH and plasma calcium concentration in some patients.

Current National Kidney Foundation Kidney Disease Outcomes Quality Initiative guidelines recommend a target PTH level between 2 and 9 times the upper limit of normal, recognizing that very low PTH levels are associated with adynamic bone disease and possible consequences of fracture and ectopic calcification.

CARDIOVASCULAR ABNORMALITIES

Cardiovascular disease is the leading cause of morbidity and mortality in patients at every stage of CKD. The incremental risk of cardiovascular disease in those with CKD compared to the age- and sex-matched general population ranges from 10- to 200-fold, depending on the stage of CKD. As a result, most patients with CKD succumb to cardiovascular disease **(Fig. 311-6)** before ever reaching stage 5 CKD. Between 30 and 45% of those patients who do reach stage 5 CKD have advanced significant cardiovascular complications. Thus, the focus of patient care in earlier CKD stages should be directed to prevention of cardiovascular complications.

Vascular Disease The increased prevalence of vascular disease in CKD patients derives from both traditional ("classic") and nontraditional (CKD-related) risk factors. Traditional risk factors include hypertension, diabetes mellitus, hypervolemia, dyslipidemia, sympathetic overactivity, and hyperhomocysteinemia. The CKD-related risk factors comprise anemia, hyperphosphatemia, hyperparathyroidism, increased FGF-23, sleep apnea, and systemic inflammation. The inflammatory state appears to accelerate vascular occlusive disease, and low levels of fetuin may permit more rapid vascular calcification, especially in the face of hyperphosphatemia. Other abnormalities seen in CKD may augment myocardial ischemia, including left ventricular hypertrophy and microvascular disease. In addition, hemodialysis, with its attendant episodes of hypotension and hypovolemia, may further aggravate coronary ischemia and repeatedly stun the myocardium. Interestingly, however, the largest increment in cardiovascular mortality rate in dialysis patients is not necessarily directly associated with acute myocardial infarction but, instead, is the result of congestive heart failure and sudden death. ECG monitoring studies have suggested that asystole and bradyarrhythmias are the principal causes of sudden cardiac death in dialysis patients.

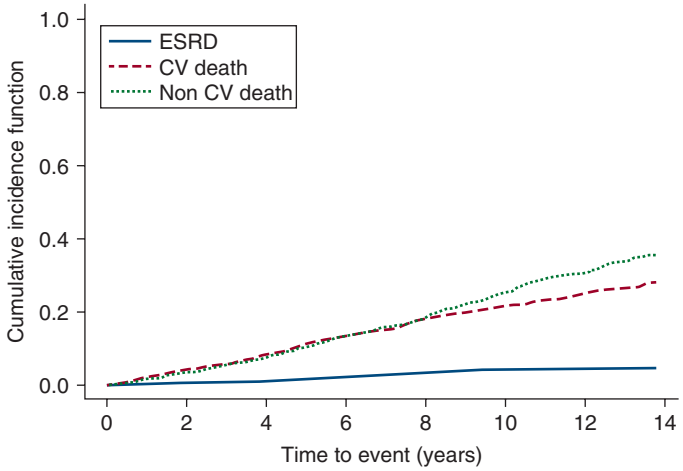


FIGURE 311-6 The cumulative incidence of end-stage renal disease (ESRD), cardiovascular (CV) death, and non-CV death during follow-up in cohort of 1268 participants with an estimated glomerular filtration rate (eGFR). *(Reproduced with permission from LS Dalrymple et al: Chronic kidney disease and the risk of end-stage renal disease versus death. J Gen Int Med 26:379, 2010.)*

Cardiac troponin levels are frequently elevated in CKD without evidence of acute ischemia. The elevation complicates the diagnosis of acute myocardial infarction in this population. Serial measurements may be needed. Therefore, the trend in levels over the hours after presentation may be more informative than a single, elevated level. Interestingly, consistently elevated levels are an independent prognostic factor for adverse cardiovascular events.

Heart Failure Abnormal cardiac function secondary to myocardial ischemia, left ventricular hypertrophy, diastolic dysfunction, and frank cardiomyopathy, in combination with salt and water retention, often results in heart failure or even pulmonary edema. Heart failure can be a consequence of diastolic or systolic dysfunction, or both. A form of "low-pressure" pulmonary edema can also occur in advanced CKD, manifesting as shortness of breath and a "bat wing" distribution of alveolar edema fluid on chest x-ray. This finding can occur even in the absence of ECFV overload and is associated with normal or mildly elevated pulmonary capillary wedge pressure. This process has been ascribed to increased permeability of alveolar capillary membranes as a manifestation of the uremic state, and it responds to dialysis. Other CKD-related risk factors, including anemia and sleep apnea, may contribute to the risk of heart failure.

Hypertension and left ventricular hypertrophy are the most common complications of CKD. Hypertension usually develops early during the course of CKD and is associated with adverse outcomes, including the development of ventricular hypertrophy and a more rapid loss of renal function. Left ventricular hypertrophy and dilated cardiomyopathy are among the strongest risk factors for cardiovascular morbidity and mortality in patients with CKD and are thought to be related primarily, but not exclusively, to prolonged hypertension and ECFV overload. In addition, anemia and the placement of an arteriovenous fistula for hemodialysis can generate a high cardiac output state and consequent high-output heart failure.

The absence of hypertension may signify poor left ventricular function. Indeed, in epidemiologic studies of dialysis patients, low blood pressure actually carries a worse prognosis than does high blood pressure. This mechanism, in part, accounts for the "reverse causation" seen in dialysis patients, wherein the presence of traditional risk factors, such as hypertension, hyperlipidemia, and obesity, appear to portend a better prognosis. Importantly, these observations derive from cross-sectional studies of late-stage CKD patients and should not be interpreted to discourage appropriate management of these risk factors in CKD patients, especially at early stages. In contrast to the general population, it is possible that in late-stage CKD, low blood pressure, reduced body mass index, and hypolipidemia indicate the presence of an advanced malnutrition-inflammation state, with the attendant poor prognosis.

The use of exogenous ESAs can increase blood pressure and the requirement for antihypertensive drugs. Chronic ECFV overload is also a contributor to hypertension, and improvement in blood pressure can often be seen with the use of dietary sodium restriction, diuretics, and fluid removal with dialysis. Nevertheless, because of activation of the RAS and other disturbances in the balance of vasoconstrictors and vasodilators, some patients remain hypertensive, despite careful attention to ECFV status.

TREATMENT

Cardiovascular Abnormalities

MANAGEMENT OF HYPERTENSION

The overarching goal of hypertension therapy in CKD is to prevent the extrarenal complications of high blood pressure, such as cardiovascular disease and stroke. Although a clear-cut generalizable benefit in slowing progression of CKD remains as yet unproven, the benefit for cardiac and cerebrovascular health is compelling. In all patients with CKD, blood pressure should be controlled to levels recommended by national guideline panels. In CKD patients with diabetes or proteinuria >1 g per 24 h, blood pressure should

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be reduced to <130/80 mmHg, if achievable without prohibitive adverse effects. Salt restriction should be the first line of therapy. When volume management alone is not sufficient, the choice of antihypertensive agent is similar to that in the general population. Angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) appear to slow the rate of decline of kidney function in a manner that extends beyond reduction of systemic arterial pressure and that involves reduction in the intraglomerular hyperfiltration and hypertension. Occasionally, introduction of ACE inhibitors and ARBs can actually precipitate an episode of AKI, especially when used in combination in patients with ischemic renovascular disease.

Slight reduction of GFR (<30% of baseline) may signify a salutary reduction in intraglomerular hypertension and hyperfiltration, and, if stable over time, can be tolerated with continued monitoring. Progressive decline in GFR should prompt discontinuation of these agents. The use of ACE inhibitors and ARBs may also be complicated by the development of hyperkalemia. Often the concomitant use of a combination of kaliuretic diuretics (e.g., furosemide with metolazone) or a potassium-lowering GI tract binder, such as patiromer, can improve potassium excretion in addition to improving blood pressure control. Potassium-sparing diuretics, such as amiloride and triamterene, should be avoided in most patients, and mineralocorticoid receptor blockers should also be used with great caution and with careful monitoring of serum potassium concentration, weighing potential cardiovascular benefits against risk for lethal hyperkalemia.

The recent movement to even lower blood pressure targets in the general population may not be applicable to patients with CKD, who often lack autoregulation to maintain GFR in the face of low perfusion pressure. If a patient experiences sudden decline in kidney function with intensification of antihypertensive therapy, consideration should be given to reducing therapy.

MANAGEMENT OF CARDIOVASCULAR DISEASE

There are many strategies available to treat the traditional and nontraditional risk factors in CKD patients. Although these have proved effective in the general population, there is little evidence for their benefit in patients with advanced CKD, especially those on dialysis. Certainly, hypertension and dyslipidemia promote atherosclerotic disease and are treatable complications of CKD. Renal disease complicated by nephrotic syndrome is associated with a very atherogenic lipid profile and hypercoagulability, which increases the risk of occlusive vascular disease. Because diabetes mellitus and hypertension are the two most frequent causes of advanced CKD, it is not surprising that cardiovascular disease is the most frequent cause of death in dialysis patients. The use of the gliflozins (SGLT2 inhibitors) in patients with diabetes mellitus has recently been associated with kidney protection and a reduction in cardiovascular events, including heart failure. Currently under study is the feasibility of using gliflozins in nondiabetic CKD.

The role of "inflammation" may be quantitatively more important in patients with kidney disease, and the treatment of more traditional risk factors may result in only modest success. However, modulation of traditional risk factors may be the only weapon in the therapeutic armamentarium for these patients until the nature of inflammation in CKD and its treatment are better understood.

Pericardial Disease Chest pain with respiratory accentuation, accompanied by a friction rub, is diagnostic of pericarditis. Classic electrocardiographic abnormalities include PR-interval depression and diffuse ST-segment elevation. Pericarditis can be accompanied by pericardial effusion that is seen on echocardiography and can rarely lead to tamponade. However, the pericardial effusion can be asymptomatic, and pericarditis can be seen without significant effusion.

Pericarditis is observed in advanced uremia and, with the advent of timely initiation of dialysis, is not as common as it once was. It is now more often observed in underdialyzed, nonadherent patients than in those starting dialysis.

TABLE 311-5 Causes of Anemia in Chronic Kidney Disease (CKD)

Relative deficiency of erythropoietin
Diminished red blood cell survival
Bleeding diathesis
Iron deficiency due to poor dietary absorption and gastrointestinal blood loss
Hyperparathyroidism/bone marrow fibrosis
Chronic inflammation
Folate or vitamin B_{12} deficiency
Hemoglobinopathy
Comorbid conditions: hypo-/hyperthyroidism, pregnancy, HIV-associated disease, autoimmune disease, immunosuppressive drugs

TREATMENT

Pericardial Disease

Uremic pericarditis is an absolute indication for the urgent initiation of dialysis or for intensification of the dialysis prescription in those already receiving dialysis. Because of the propensity to hemorrhage in pericardial fluid, hemodialysis should be performed without heparin. A pericardial drainage procedure should be considered in patients with recurrent pericardial effusion, especially with echocardiographic signs of impending tamponade. Nonuremic causes of pericarditis and effusion include viral, malignant, tuberculous, and autoimmune etiologies. It may also be seen after myocardial infarction and as a complication of treatment with the antihypertensive drug minoxidil. Consideration could be given to the use of colchicine or nonsteroidal anti-inflammatory drugs, although the latter agents could adversely affect renal function.

HEMATOLOGIC ABNORMALITIES

Anemia A normocytic, normochromic anemia is observed as early as stage 3 CKD and is almost universal by stage 4. The primary cause is insufficient production of erythropoietin (EPO) by the diseased kidneys. Additional factors are reviewed in **Table 311-5**.

The anemia of CKD is associated with a number of adverse pathophysiologic consequences, including decreased tissue oxygen delivery and utilization, increased cardiac output, ventricular dilation, and ventricular hypertrophy. Clinical manifestations include fatigue and diminished exercise tolerance, angina, heart failure, decreased cognition and mental acuity, and impaired host defense against infection. In addition, anemia may play a role in growth restriction in children with CKD. Although many studies in CKD patients have found that anemia and resistance to exogenous ESAs are associated with a poor prognosis, the relative contribution to a poor outcome of the low hematocrit itself, versus inflammation as a cause of the anemia and ESA resistance, remains unclear.

TREATMENT

Anemia

The availability of recombinant human ESA has been one of the most significant advances in the care of renal patients since the introduction of dialysis and renal transplantation. Its routine use has obviated the need for regular blood transfusions in severely anemic CKD patients, thus dramatically reducing the incidence of transfusion-associated infections and iron overload.

Frequent blood transfusions in dialysis patients also lead to the development of alloantibodies that can sensitize the patient to donor kidney antigens and make kidney transplantation more problematic.

Adequate bone marrow iron stores should be available before treatment with ESA is initiated. Iron supplementation is usually essential to ensure an optimal response to ESA in patients with CKD because the demand for iron by the marrow frequently

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exceeds the amount of iron that is immediately available for erythropoiesis (measured by percent transferrin saturation), as well as the amount in iron stores (measured by serum ferritin). For the CKD patient not yet on dialysis or the patient treated with peritoneal dialysis, oral iron supplementation should be attempted. If there is GI intolerance or poor GI absorption, the patient may have to undergo IV iron infusion. For patients on hemodialysis, IV iron can be administered during dialysis, keeping in mind that parenteral iron therapy can increase the susceptibility to bacterial infections and that the adverse effects of free serum iron are still under investigation. In addition to iron, an adequate supply of other major substrates and cofactors for red cell production must be ensured, including vitamin B_{12} and folate. Anemia resistant to recommended doses of ESA in the face of adequate iron stores may be due to some combination of the following: acute or chronic inflammation, inadequate dialysis, severe hyperparathyroidism, chronic blood loss or hemolysis, chronic infection, or malignancy.

A new class of agents to treat the anemia of CKD are the prolyl-hydroxylase inhibitors of endogenous hypoxia-inducible factors (HIFs). This inhibition leads to an increase in both endogenous production of EPO and an increase in GI absorption of iron. Studies are in progress comparing the efficacy of these agents to the standard ESAs.

Randomized, controlled trials of ESA in CKD have failed to show an improvement in cardiovascular outcomes with this therapy. Indeed, there has been an indication that the use of ESA in CKD may be associated with an increased risk of stroke in those with type 2 diabetes or an increase in thromboembolic events and perhaps a faster progression of renal decline.

Therefore, any benefit in terms of improvement of anemic symptoms needs to be balanced against the potential cardiovascular risk. Although further studies are needed, it is quite clear that normalization of the hemoglobin concentration has not been demonstrated to be of incremental benefit to CKD patients. Current practice is to target a hemoglobin concentration of 100–115 g/L.

Abnormal Hemostasis Patients with later stages of CKD may have a prolonged bleeding time, decreased activity of platelet factor III, abnormal platelet aggregation and adhesiveness, and impaired prothrombin consumption. Clinical manifestations include an increased tendency to bleeding and bruising, prolonged bleeding from surgical incisions, menorrhagia, and GI bleeding. Interestingly, CKD patients also have a greater susceptibility to thromboembolism, especially if they have renal disease that includes nephrotic-range proteinuria. The latter condition results in hypoalbuminemia and renal loss of anticoagulant factors, which can lead to a thrombophilic state.

TREATMENT

Abnormal Hemostasis

Abnormal bleeding time and coagulopathy in patients with renal failure may be reversed temporarily with desmopressin (DDAVP), cryoprecipitate, IV conjugated estrogens, blood transfusions, and ESA therapy. Optimal dialysis will usually correct a prolonged bleeding time.

Given the coexistence of bleeding disorders and a propensity to thrombosis that is unique in the CKD patient, decisions about anticoagulation that have a favorable risk-benefit profile in the general population may not be applicable to the patient with advanced CKD. One example is warfarin anticoagulation for atrial fibrillation; the decision to anticoagulate should be made on an individual basis in the CKD patient because there appears to be a greater risk of bleeding complications.

Certain anticoagulants, such as fractionated low-molecular-weight heparin, may need to be avoided or dose-adjusted in these patients, with monitoring of factor Xa activity where available. It is often more prudent to use conventional unfractionated heparin, titrated to the measured partial thromboplastin time, in hospitalized patients requiring an alternative to warfarin anticoagulation. The new classes of oral anticoagulants are all, in part, renally eliminated and need to be avoided or dose adjusted in the face of decreased GFR (Chap. 118).

NEUROMUSCULAR ABNORMALITIES

Central nervous system (CNS), peripheral, and autonomic neuropathy, as well as abnormalities in muscle structure and function, are all well-recognized complications of CKD. Subtle clinical manifestations of uremic neuromuscular disease usually become evident at stage 3 CKD.

Early manifestations of CNS complications include mild disturbances in memory and disturbances in concentration and sleep. Neuromuscular irritability, including hiccups, cramps, and twitching, becomes evident at later stages. In advanced untreated kidney failure, asterixis, myoclonus, seizures, and coma can be seen.

Peripheral neuropathy usually becomes clinically evident after the patient reaches stage 4 CKD, although electrophysiologic and histologic evidence occurs earlier. Initially, sensory nerves are involved more than motor, lower extremities more than upper, and distal parts of the extremities more than proximal. The "restless leg syndrome" is characterized by ill-defined sensations of sometimes debilitating discomfort in the legs and feet relieved by frequent leg movement. Evidence of peripheral neuropathy without another cause (e.g., diabetes mellitus or iron deficiency) is an indication for starting renal replacement therapy. Many of the complications described above will resolve with dialysis, although subtle nonspecific abnormalities may persist.

GASTROINTESTINAL AND NUTRITIONAL ABNORMALITIES

Uremic fetor, a urine-like odor on the breath, derives from the breakdown of urea to ammonia in saliva and is often associated with an unpleasant metallic taste (dysgeusia). Gastritis, peptic disease, and mucosal ulcerations at any level of the GI tract occur in uremic patients and can lead to abdominal pain, nausea, vomiting, and GI bleeding. These patients are also prone to constipation, which can be worsened by the administration of calcium and iron supplements. The retention of uremic toxins also leads to anorexia, nausea, and vomiting.

Protein restriction may be useful to decrease nausea and vomiting; however, it may put the patient at risk for malnutrition and should be carried out, if possible, in consultation with a registered dietitian specializing in the management of CKD patients. Weight loss and protein-energy malnutrition, consequences of low protein and caloric intake, are common in advanced CKD and are often an indication for initiation of renal replacement therapy. Metabolic acidosis and the activation of inflammatory cytokines can promote protein catabolism. A number of indices are useful in nutritional assessment and include dietary history, including food diary, and subjective global assessment; edema-free body weight; and measurement of urinary protein nitrogen appearance. Dual-energy x-ray absorptiometry bioimpedance analysis is now widely used to estimate lean body mass versus fluid weight. Nutritional guidelines for patients with CKD are summarized in the "Treatment" section.

ENDOCRINE-METABOLIC DISTURBANCES

Glucose metabolism is impaired in CKD. However, fasting blood glucose is usually normal or only slightly elevated, and mild glucose intolerance does not require specific therapy. Because the kidney contributes to insulin removal from the circulation, plasma levels of insulin are slightly to moderately elevated in most uremic patients, both in the fasting and postprandial states. Because of this diminished renal degradation of insulin, patients on insulin therapy may need progressive reduction in dose as their renal function worsens. Many anti-hyperglycemic agents, including the gliptins, require dose reduction in renal failure, and some, such as metformin and sulfonylureas, are contraindicated when the GFR is less than half of normal. The gliflozins, discussed above, that inhibit sodium-glucose transport in the proximal tubule result in glucose lowering, accompanied by striking reductions in kidney function decline and in cardiovascular events.

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The stabilization of GFR in many patients with this therapeutic intervention represents a major, important added beneficial effect of these drugs. Their long-term stabilizing effect on GFR and urine albumin excretion appears to result from correction of hyperfiltration early in type 2 diabetes mellitus via reactivation of the tubuloglomerular feedback loop. This represents a fortunate convergence of pathophysiology of glomerular hyperfiltration in diabetes with drug discovery. A similar effect on hyperfiltration by residual nephrons in certain nondiabetic forms of CKD may explain the salutary role of this class of medications more broadly in CKD. Other studies have also pointed to a more direct effect on proximal tubule metabolic pathways that alleviate cell injury.

In women with CKD, estrogen levels are low, and menstrual abnormalities, infertility, and inability to carry pregnancies to term are common. When the GFR has declined to ~40 mL/min, pregnancy is associated with a high rate of spontaneous abortion, with only ~20% of pregnancies leading to live births, and pregnancy may hasten the progression of the kidney disease itself. Women with CKD who are contemplating pregnancy should consult first with a nephrologist in conjunction with an obstetrician specializing in high-risk pregnancy. Men with CKD have reduced plasma testosterone levels, and sexual dysfunction and oligospermia may supervene. Sexual maturation may be delayed or impaired in adolescent children with CKD, even among those treated with dialysis. Many of these abnormalities improve or reverse with intensive dialysis or with successful renal transplantation.

DERMATOLOGIC ABNORMALITIES

Abnormalities of the skin are prevalent in progressive CKD. Pruritus is quite common and one of the most vexing manifestations of the uremic state. In advanced CKD, even on dialysis, patients may become more pigmented, and this is felt to reflect the deposition of retained pigmented metabolites, or urochromes. Although many of the cutaneous abnormalities improve with dialysis, pruritus is often tenacious. The first lines of management are to rule out unrelated skin disorders, such as scabies, and to treat hyperphosphatemia, which can cause itch. Local moisturizers, mild topical glucocorticoids, oral antihistamines, and ultraviolet radiation have been reported to be helpful. Recently, agonists of kappa opioid receptors have shown promise in reducing pruritis in hemodialysis patients.

A skin condition unique to CKD patients called nephrogenic fibrosing dermopathy consists of progressive subcutaneous induration, especially on the arms and legs. The condition is seen very rarely in patients with CKD who have been exposed to the magnetic resonance contrast agent gadolinium. Current recommendations are that patients with CKD stage 3 (GFR 30–59 mL/min) should minimize exposure to gadolinium and those with CKD stages 4–5 (GFR <30 mL/min) should avoid the use of gadolinium agents unless it is medically necessary. However, no patient should be denied an imaging investigation that is critical to management, and under such circumstances, rapid removal of gadolinium by hemodialysis (even in patients not yet receiving renal replacement therapy) shortly after the procedure may mitigate this sometimes devastating complication.

EVALUATION AND MANAGEMENT OF PATIENTS WITH CKD

INITIAL APPROACH

History and Physical Examination Symptoms and overt signs of kidney disease are often subtle or absent until renal failure supervenes. Thus, the diagnosis of kidney disease often surprises patients and may be a cause of skepticism and denial. Particular aspects of the history that are germane to renal disease include a history of hypertension (which can cause CKD or more commonly be a consequence of CKD), diabetes mellitus, abnormal urinalyses, and problems with pregnancy such as preeclampsia or early pregnancy loss. A careful drug history should be elicited. Drugs to consider include nonsteroidal anti-inflammatory agents, cyclooxygenase-2 (COX-2) inhibitors, antimicrobials, chemotherapeutic agents, antiretroviral agents, proton pump inhibitors, phosphate-containing bowel cathartics, and lithium. In evaluating the uremic syndrome, questions about appetite, weight loss, nausea, hiccups, peripheral edema, muscle cramps, pruritus, and restless legs are especially helpful. A family history of kidney disease, together with assessment of manifestations in other organ systems such as auditory, visual, and integumentary, may lead to the diagnosis of a heritable form of CKD (e.g., Alport's or Fabry's disease, cystinosis) or shared environmental exposure to nephrotoxic agents (e.g., heavy metals, aristolochic acid). It should be noted that clustering of CKD, sometimes of different etiologies, is often observed within families.

The physical examination should focus on blood pressure and target organ damage from hypertension. Thus, funduscopy and precordial examination should be carried out. Funduscopy is especially important in the diabetic patient, because it may show evidence of diabetic retinopathy, which is associated with diabetic nephropathy. Other physical examination manifestations of CKD include edema and sensory polyneuropathy. The finding of asterixis or a pericardial friction rub not attributable to other causes usually signifies the presence of the uremic syndrome.

Laboratory Investigation Laboratory studies should focus on a search for clues to an underlying causative or aggravating disease process and on the degree of renal damage and its consequences. Serum and urine protein electrophoresis, looking for multiple myeloma, should be obtained in all patients >35 years old with unexplained CKD, especially if there is associated anemia and elevated, or even inappropriately normal, serum calcium concentration in the face of renal insufficiency. In the presence of glomerulonephritis, autoimmune diseases such as lupus and underlying infectious etiologies such as hepatitis B and C and HIV should be tested. Serial measurements of renal function should be obtained to determine the pace of renal deterioration and ensure that the disease is truly chronic rather than acute or subacute and hence potentially reversible. Serum concentrations of calcium, phosphorus, vitamin D, and PTH should be measured to evaluate metabolic bone disease. Hemoglobin concentration, iron, vitamin B_{12}, and folate should also be evaluated. A 24-h urine collection may be helpful, because protein excretion >300 mg may be an indication for therapy with ACE inhibitors or ARBs and also is associated with a higher risk of progression.

Imaging Studies The most useful imaging study is a renal ultrasound, which can verify the presence of two kidneys, determine if they are symmetric, provide an estimate of kidney size, and rule out renal masses and evidence of obstruction. Because it takes time for kidneys to shrink as a result of chronic disease, the finding of bilaterally small kidneys supports the diagnosis of CKD of long-standing duration. If the kidney size is normal, it is possible that the kidney disease is acute or subacute. The exceptions are diabetic nephropathy (where kidney size is increased at the onset of diabetic nephropathy before CKD supervenes), amyloidosis, and HIV nephropathy, where kidney size may be normal in the face of CKD. Polycystic kidney disease that has reached some degree of renal failure will almost always present with enlarged kidneys with multiple cysts (Chap. 315). A discrepancy >1 cm in kidney length suggests either a unilateral developmental abnormality or a disease process or renovascular disease with arterial insufficiency affecting one kidney more than the other. The diagnosis of renovascular disease can be undertaken with different techniques, including Doppler sonography, nuclear medicine studies, or CT or magnetic resonance imaging (MRI) studies. If there is a suspicion of reflux nephropathy (recurrent childhood urinary tract infection, asymmetric renal size with scars on the renal poles), a voiding cystogram may be indicated. However, in most cases, by the time the patient has CKD, the reflux has resolved, and even if still present, repair does not improve renal function. Radiographic contrast imaging studies are not particularly helpful in the investigation of CKD. Intravenous or intraarterial dye should be avoided where possible in the CKD patient, especially with diabetic nephropathy, because of the risk of radiographic contrast dye–induced renal failure. When unavoidable, appropriate precautionary measures include avoidance of hypovolemia at the time of contrast exposure, minimization of the dye load, and choice of radiographic contrast preparations with the least nephrotoxic potential. Additional measures thought to attenuate contrast-induced worsening of renal

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function include judicious administration of sodium bicarbonate–containing solutions and N-acetylcysteine, although these agents may not be as effective as previously thought.

Kidney Biopsy In the patient with bilaterally small kidneys, renal biopsy is not advised because (1) it is technically difficult and has a greater likelihood of causing bleeding and other adverse consequences, (2) there is usually so much scarring that the underlying disease may not be apparent, and (3) the window of opportunity to render disease-specific therapy has passed. Other contraindications to renal biopsy include uncontrolled hypertension, active urinary tract infection, bleeding diathesis (including ongoing anticoagulation), and severe obesity. Ultrasound-guided percutaneous biopsy is the favored approach, but a surgical or laparoscopic approach can be considered, especially in the patient with a single kidney where direct visualization and control of bleeding are crucial. In the CKD patient in whom a kidney biopsy is indicated (e.g., suspicion of a concomitant or superimposed active process such as interstitial nephritis or in the face of accelerated loss of GFR), the bleeding time should be measured, and if increased, desmopressin should be administered immediately prior to the procedure.

A brief run of hemodialysis (without heparin) may also be considered prior to renal biopsy to normalize the bleeding time.

ESTABLISHING THE DIAGNOSIS AND ETIOLOGY OF CKD

The most important initial diagnostic step is to distinguish newly diagnosed CKD from acute or subacute renal failure, because the latter two conditions may respond to targeted therapy. Previous measurements of serum creatinine concentration are particularly helpful in this regard. Normal values from recent months or even years suggest that the current extent of renal dysfunction could be more acute, and hence reversible, than might otherwise be appreciated. In contrast, elevated serum creatinine concentration in the past suggests that the renal disease represents a chronic process. Even if there is evidence of chronicity, there is the possibility of a superimposed acute process (e.g., ECFV depletion, urinary infection or obstruction, or nephrotoxin exposure) supervening on the chronic condition. If the history suggests multiple systemic manifestations of recent onset (e.g., fever, polyarthritis, rash), it should be assumed that renal insufficiency is part of an acute systemic illness.

Although kidney biopsy can usually be performed in early CKD (stages 1–3), it is not always indicated. For example, in a patient with a history of type 1 diabetes mellitus for 15–20 years with retinopathy, nephrotic-range proteinuria, and absence of hematuria, the diagnosis of diabetic nephropathy is very likely and biopsy is usually not necessary. However, if there is another finding not typical of diabetic nephropathy, such as hematuria or white blood cell casts, or absence of diabetic retinopathy, some other disease may be present and a biopsy may be indicated.

In the absence of a clinical diagnosis, kidney biopsy may be the only recourse to establish an etiology in early-stage CKD. However, as noted above, once the CKD is advanced and the kidneys are small and scarred, there is little utility and significant risk in attempting to arrive at a specific diagnosis. Genetic testing using a combination of chromosomal microarray and whole exome sequencing is increasingly entering the repertoire of diagnostic tests since the patterns of injury and kidney morphologic abnormalities often reflect overlapping causal mechanisms, whose origins can sometimes be attributed to a genetic predisposition or cause (Table 311-2).

TREATMENT

Chronic Kidney Disease

Treatments aimed at specific causes of CKD are discussed elsewhere. Two recent developments in the etiology-directed therapy of CKD include the now-proven role of gliflozins in diabetic kidney disease and the emergence of genome-specific therapies now established for certain patients with ADPKD **(Chap. 315)**, which are at the clinical trial stage for *APOL1*-mediated kidney disease and certain forms of hyperoxaluria. The optimal timing of both specific and nonspecific therapy is usually well before there has been a measurable decline in GFR and certainly before CKD is established. It is helpful to measure sequentially and plot the rate of decline of GFR in all patients. Any acceleration in the rate of decline should prompt a search for superimposed acute or subacute processes that may be reversible. These include ECFV depletion, uncontrolled hypertension, urinary tract infection, new obstructive uropathy, exposure to nephrotoxic agents (such as nonsteroidal anti-inflammatory drugs [NSAIDs] or radiographic dye), and reactivation or flare of the original disease, such as lupus or vasculitis.

SLOWING THE PROGRESSION OF CKD

There is variation in the rate of decline of GFR among patients with CKD. However, the following interventions should be considered in an effort to stabilize or slow the decline of renal function.

Reducing Intraglomerular Hypertension and Proteinuria Increased intraglomerular filtration pressures and glomerular hypertrophy develop as a response to loss of nephron number. This response is maladaptive as it promotes the ongoing decline of kidney function even if the inciting process has been treated or spontaneously resolved. Control of glomerular hypertension is important in slowing the progression of CKD. Moreover, elevated blood pressure increases proteinuria by increasing its flux across the glomerular capillaries. Conversely, the renoprotective effect of antihypertensive medications is gauged through the consequent reduction of proteinuria. Thus, the more effective a given treatment is in lowering protein excretion, the greater is the subsequent impact on protection from decline in GFR. This observation is the basis for the treatment guideline establishing 130/80 mmHg as a target blood pressure in proteinuric CKD patients.

Several controlled studies have shown that ACE inhibitors and ARBs are effective in slowing the progression of renal failure in patients with advanced stages of both diabetic and nondiabetic CKD, in large part through effects on efferent vasodilatation and the subsequent decline in glomerular hypertension. In the absence of an anti-proteinuric response with either agent alone, combined treatment with both ACE inhibitors and ARBs has been considered. The combination is associated with a greater reduction in proteinuria compared to either agent alone. Insofar as reduction in proteinuria is a surrogate for improved renal outcome, the combination would appear to be advantageous. However, there is a greater incidence of AKI and adverse cardiac events from such combination therapy. On balance, therefore, ACE inhibitor plus ARB therapy should be avoided. A progressive increase in serum creatinine concentration with these agents may suggest the presence of renovascular disease within the large or small arteries.

Among the calcium channel blockers, diltiazem and verapamil may exhibit superior antiproteinuric and renoprotective effects compared to the dihydropyridines. At least two different categories of response can be considered: one in which progression is strongly associated with systemic and intraglomerular hypertension and proteinuria (e.g., diabetic nephropathy, glomerular diseases) and in which ACE inhibitors and ARBs are recommended choices, and another in which proteinuria is mild or absent initially (e.g., ADPKD and other tubulointerstitial diseases), where the contribution of intraglomerular hypertension is less prominent and other antihypertensive agents can be useful for control of systemic hypertension.

MANAGING OTHER COMPLICATIONS OF CKD

Medication Dose Adjustment Although the loading dose of most drugs is not affected by CKD because renal elimination is not used in the calculation, the maintenance doses of many drugs will need to be adjusted. For those agents in which >70% excretion is by a nonrenal route, such as hepatic elimination, dose adjustment may not be needed. Some drugs that should be avoided include

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metformin, meperidine, and oral anti-hyperglycemics that are eliminated by the kidney. NSAIDs should be avoided because of the risk of further worsening of kidney function. Many antibiotics, antihypertensives, and antiarrhythmics may require a reduction in dosage or change in the dose interval. Several online Web-based databases for dose adjustment of medications according to stage of CKD or estimated GFR are available (e.g., *http://www.globalrph.com/index_renal.htm*). Nephrotoxic radiocontrast agents and gadolinium should be avoided or used according to strict guidelines when medically necessary, as discussed above.

PREPARATION FOR RENAL REPLACEMENT THERAPY

(See also Chap. 313) Temporary relief of symptoms and signs of impending uremia, such as anorexia, nausea, vomiting, lassitude, and pruritus, may sometimes be achieved with dietary protein restriction. However, this diet carries a risk of malnutrition; thus, plans for more long-term management should be in place.

Maintenance dialysis and kidney transplantation have extended the lives of hundreds of thousands of patients with CKD worldwide. Clear indications for initiation of renal replacement therapy for patients with CKD include anorexia and nausea not attributable to reversible causes such as peptic ulcer disease, evidence of malnutrition, and fluid and electrolyte abnormalities, principally hyperkalemia or ECFV overload, that are refractory to other measures. Encephalopathy and pericarditis are very late complications, so it is now rare that they serve as indications for initiation of renal replacement therapy.

Recommendations for the Optimal Time for Initiation of Renal Replacement Therapy Because of the individual variability in the severity of uremic symptoms and renal function, it is ill-advised to assign an arbitrary urea nitrogen or creatinine level to the need to start dialysis. Moreover, patients may become accustomed to chronic uremia and deny symptoms, only to find that they feel better with dialysis and realize in retrospect how poorly they were feeling before its initiation.

Previous studies suggested that starting dialysis before the onset of severe symptoms and signs of uremia was associated with prolongation of survival. This led to the concept of "healthy" start and is congruent with the philosophy that it is better to keep patients feeling well rather than allowing them to become ill with uremia and then attempting to return them to better health with dialysis or transplantation. Although recent studies have not confirmed an association of early-start dialysis with improved patient survival, there may be merit in this approach for some patients. On a practical level, advanced preparation may help to avoid problems with the dialysis process itself (e.g., a poorly functioning fistula for hemodialysis or malfunctioning peritoneal dialysis catheter) and, thus, preempt the morbidity associated with resorting to the insertion of temporary hemodialysis access with its attendant risks of sepsis, bleeding, thrombosis, and association with accelerated mortality.

Patient Education Social, psychological, and physical preparation for the transition to renal replacement therapy and the choice of the optimal initial modality are best accomplished with a gradual approach involving a multidisciplinary team. Along with conservative measures discussed in the sections above, it is important to prepare patients with an intensive educational program, explaining the likelihood and timing of initiation of renal replacement therapy and the various forms of therapy available and the option of nondialytic conservative care. The more knowledgeable that patients are about hemodialysis (both in-center and home-based), peritoneal dialysis, and kidney transplantation, the easier and more appropriate will be their decisions. Patients who are provided with education are more likely to choose home-based dialysis therapy. This approach is of societal benefit because home-based therapy is less expensive to most jurisdictions and is associated with improved quality of life. The educational programs should be commenced no later than stage 4 CKD so that the patient has sufficient time and cognitive function to learn the important concepts, make informed choices, and implement preparatory measures for renal replacement therapy.

Exploration of social support is also important. Early education of family members for selection and preparation of a home dialysis helper or a biologically or emotionally related potential living kidney donor should occur long before the onset of symptomatic renal failure.

Kidney transplantation **(Chap. 313)** offers the best potential for complete rehabilitation because dialysis replaces only a small fraction of the kidneys' filtration function and none of the other renal functions, including endocrine and anti-inflammatory effects. Generally, kidney transplantation follows a period of dialysis treatment, although preemptive kidney transplantation (usually from a living donor) can be carried out if it is certain that the renal failure is irreversible.

IMPLICATIONS FOR GLOBAL HEALTH

In contrast to the natural decline and successful eradication of many devastating infectious diseases, there is rapid growth in the prevalence of metabolic and vascular disease in developing countries. Diabetes mellitus is becoming increasingly prevalent in these countries, perhaps due in part to change in dietary habits, diminished physical activity, and weight gain. Therefore, it follows that there will be a proportionate increase in vascular and renal disease. Health care agencies must plan for improved screening of high-risk individuals for early detection, prevention, and treatment plans in these nations and must start considering options for improved availability of renal replacement therapies.

There is also increasing recognition of endemic nephropathies in developing countries that particularly target young males working in agriculture. The extent of morbidity and mortality associated with these nephropathies is only starting to be appreciated. It is unclear what the cause is, but population genetic risk, endemic nephrotoxins, exposure to pesticides, NSAID use, and chronic volume depletion have all been suggested to contribute.

FURTHER READING

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312 Dialysis in the Treatment of Kidney Failure

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Dialysis may be required for the treatment of either acute or chronic kidney disease (CKD). The use of continuous renal replacement therapies (CRRTs) and prolonged intermittent renal replacement therapy (PIRRT)/slow low-efficiency dialysis (SLED) is specific to the management of acute renal failure and is discussed in **Chap. 310**. These modalities are performed continuously (CRRT) or over 6–12 h per session